

DAMA | Finalized papers

Portrait A4 research comparison • 10 papers • Updated 2 October 2026

1. Comparison table

Read each result as baseline to proposed method. The baseline is named in the results column. Models are the paper's evaluated models, not DAMA model selections. One continuous table follows across pages.

NR = not reported in the cited experiment. **N/A** = no original system benchmark. **pp** = percentage points. Results are author-reported, not local reproductions. Percentage reductions are calculated from the displayed values. Compare within each row; different rows use different workloads and measurement boundaries.

Paper / method	Tested models and setup	Reported results and named baseline	Short limitation
[1] MemRouter 2026 Learned write gate	Answer: Qwen2.5-7B. Memory: frozen Qwen2.5-7B + ~12M trainable parameters; BGE-large-en-v1.5 encoder (335M). Test: LoCoMo; ~1,300 questions.	Baseline: LLM memory manager. Write p50: 970 to 58 ms/turn (94.0% lower). Write p95: 1,415 to 71 ms (95.0% lower). Total memory time: 5,028 to 293 s (94.2% lower). QA p50: 198 to 141 ms. QA throughput: 0.24 to 2.58/s. F1: 45.6 to 52.0 (+6.4 points). GPU peak: 26.6 to 31.2 GB. Retention: ~62% of turns kept; this is not a token-saving rate. Dollar cost: NR.	ADD/NOOP only. Frozen 7B backbone still required; GPU use rises. Not a complete memory lifecycle.
[2] SAGE 2026 Statistical novelty gate	Answer / memory LLM: GPT-4o-mini in this comparison. Gate: statistical novelty test, not a separate generative model. Test: full LoCoMo; Mem0 integration.	Baseline: Mem0 without SAGE. Write API cost: \$1.24 to \$0.36 (71.0% lower). Write tokens: 5,551,631 to 2,155,207 (61.2% fewer). Mean add call: 5.38 to 1.76 s (67.3% lower). Ingestion: 39.3 to 15.7 min. Search phase: 66.7 to 65.5 min. Total run: 106.3 to 81.5 min (23.3% lower). LLM calls: 3,217 to 2,297. Judge score: 53.5 to 52.2; temporal subset 42.7 to 35.4.	Novelty is not usefulness. Temporal quality falls. Dollar savings cover writes, not the full assistant.
[3] Gated-Memory Routing 2026 Within-task multi-agent memory	Model pool: Llama 3.2-3B; Llama 3.1-8B; Mistral-NeMo 12B; Qwen2.5-14B/32B. Tests: MATH, GSM-Hard, MBPP, HumanEval, MMLU-Pro.	Baseline: Puppeteer-32B. Mean accuracy: 75.29% to 77.73% (+2.44 pp). HumanEval pass@1: 85.16% to 89.84%. HumanEval cost proxy: 0.047 to 0.032 (31.9% lower). HumanEval time: 3.0 to 1.5 s/query (50% lower). MBPP time: 3.2 to 2.6 s/query. Compute: 0.144 to 0.107 PFLOPs on HumanEval. Versus MASRouter: HumanEval proxy 0.057 to 0.032 (43.9% lower). Raw tokens / dollars: NR.	Time is batch-amortized. Cost is a model-size proxy, not USD. Does not test cross-session memory.

Paper / method	Tested models and setup	Reported results and named baseline	Short limitation
[4] Memory fabric 2026 Review of shared and persistent memory	Models: no single tested model. Evidence: review screened 662 records and included 27 studies.	System benchmark: none. Latency: N/A. Token efficiency: N/A. Cost reduction: N/A. Quality gain: N/A. Provides architectural and governance requirements for persistent, shared memory; it does not measure a new system's efficiency.	Conceptual review. Cannot be used as numerical evidence of a speed or cost improvement.
[5] Memoria 2025 Session summaries + weighted knowledge graph	Answer: GPT-4.1-mini. Embedding: text-embedding-ada-002; top-20 retrieval. Store: SQLite / Chroma. Test: 148 LongMemEval-S queries: 70 user-fact, 78 update.	Baseline: full context. Prompt: ~115,000 to 398 (user facts) / 400 (updates) tokens (~99.65% fewer). User-fact accuracy: 85.7% to 87.1%. Update accuracy: 78.2% to 80.8%. Batch runtime, user facts: 391 to 260 s (33.5% lower). Batch runtime, updates: 522 to 320 s (38.7% lower). Against A-MEM (OpenAI embeddings): user-fact runtime 252 to 260 s; Memoria is slower here. Dollar cost / p50 / p95: NR.	Only two query categories. Runtime is for a batch, not one answer. Offline memory construction is excluded.
[6] MemoryRepository for AI NPC 2024 Summarization and forgetting	Shown comparison: GPT-3.5-based MemoryRepository versus MemGPT. Other study models: GPT-4, ChatGLM. Test: dialogue quality at the 100th round.	Baseline: MemGPT. Retrieval: 0.831 to 0.806. Correctness: 0.713 to 0.705. Coherence: 0.874 to 0.886. Human-likeness: 0.718 to 0.861. Latency: NR. Token reduction: NR. Dollar cost: NR. The quality dimensions move in different directions; this is not an across-the-board accuracy gain.	NPC dialogue setting. Better human-likeness comes with lower retrieval and correctness scores.
[7] MemWeave 2026 Learned graph / vector retrieval routing	Answer: GPT-4o-2024-08-06. Router: DistilBERT (66M), 1,200 labeled route examples. Embedding: text-embedding-3-small. Test: LoCoMo; 7 train / 3 test conversations.	Baseline: full context. Latency: 9,870 to 420 ms (95.7% lower); router averages 12 ms. Prompt: 26,400 to 2,320 tokens (91.2% fewer). QA F1: 73.9 to 73.5 (-0.4 points). Judge score: 75.1 to 74.6. Baseline: Mem0g. Latency 1,090 to 420 ms; prompt 2,580 to 2,320; F1 68.6 to 73.5. Index: 128 to 145 MB vs Mem0g. Dollar cost: NR.	Single benchmark and trained router. Larger index than Mem0g. More prompt tokens than vector Mem0 (2,140).

Paper / method	Tested models and setup	Reported results and named baseline	Short limitation
[8] EGMW 2026 Evidence-gated memory writing	Agent / scorer: Llama3-8B-Instruct; Qwen2.5-7B-Instruct; Mistral-7B-Instruct-v0.3. Runtime: A100 80GB, FP16, vLLM. Tests: MemoryAgentBench (MAB), LoCoMo; averages across models/runs.	Baseline: FullWrite. Memory tokens, MAB: 2,840 to 1,180 (58.5% fewer). Memory tokens, LoCoMo: 3,420 to 1,350 (60.5% fewer). Accuracy: MAB 56.8% to 72.8%; LoCoMo 48.5% to 68.1%. Pollution, MAB: 41.2% to 12.3%. Net end-to-end time: ~12% lower over 40 turns vs FullWrite. Write overhead: +340 ms/turn (+23% vs extraction-only); retrieval time 18% lower. Versus RAGWrite accuracy: +6.7 pp MAB; +8.4 pp LoCoMo. Dollar cost: NR.	Three scoring calls add upfront work. Savings depend on how often memories are read after writing.
[9] SMS-RAG 2026 Structured Memory Scheduling	Main comparison: GPT-4o-mini. Further model tests: Llama 3.1-70B; Qwen3-8B. Tests: HotpotQA, MuSiQue, 2WikiQA, Legal.	Efficiency baseline: ToG 2.0 = 100. Inference tokens: 73.3 (26.7% fewer). Mean end-to-end latency: 95.2 (4.8% lower). Coverage@2: 114.3 (14.3% higher). Quality baseline: Vanilla RAG. HotpotQA F1 35.44 to 63.45; exact match 27.40 to 51.83. MuSiQue F1 23.40 to 37.20. 2WikiQA F1 42.93 to 66.48. Legal F1 32.19 to 37.88. Absolute time / tokens / USD: NR in the efficiency figure.	External evidence scheduling, not persistent personal memory. Normalized chart values are not milliseconds or token counts.
[10] ReMAP 2025 Personalized travel-planning memory	Backbone: Qwen3-235B-A22B. Test: Tibet travel scenarios; cold-start profiling and component ablations.	Personalization: 91.51%; paper reports +10.51% versus non-cold-start setup. Guide module enabled vs disabled: feasibility +12.38%; completeness +14.17%; simulated human score +9.07%; execution +12.60%. Latency: NR. Token reduction: NR. Dollar cost: NR. Gains retain the authors' percent wording; they are not relabeled as percentage points.	Travel-specific evaluation. Several components change together; gains cannot be attributed only to memory.

Source IDs [1]-[10] map to the full paper titles, links and evidence locations in Section 3. Published API costs reflect the paper's run and pricing, not a current DAMA quote.

2. Objectives for DAMA

Purpose: use a small Memory Agent to prepare a compact, current memory packet for a separate Primary Agent. DAMA is still a proposal: no implementation, local benchmark or measured saving is established in this repository.

Realistic first milestone: prove useful savings on one reproducible configuration before training a custom controller or expanding to every memory operation. The numbers below are provisional success criteria for our experiments, not predictions or claimed results.

Objective 1 | Build a working, measurable prototype

Keep one Primary Agent fixed. Use an existing small model for structured memory decisions, with SQL plus vector retrieval. Implement store, retrieve, update, a memory-token budget, source IDs and timestamps. Retain superseded facts as history. **Acceptance:** every operation passes scripted integration cases, decisions are logged, and uncertain retrieval can expand context.

Objective 2 | Preserve answer quality

Initial target: answer accuracy within 2 percentage points of the strongest reproducible baseline under the same Primary Agent. Test full LoCoMo and a preregistered LongMemEval split, including temporal and update questions. Use full context, standard RAG and an LLM memory manager as baselines. Report confidence intervals and category scores; a mean alone does not establish equivalence.

Objective 3 | Reduce tokens and total operating cost

Initial target: at least 50% fewer historical-context tokens than full context while meeting Objective 2. **Research target:** at least 20% lower total model cost than the LLM memory manager. Count memory-model calls, Primary Agent calls, embeddings, writes, retries and maintenance. Log measured API charges or a stated pricing calculation; for local models, report runtime and hardware use separately.

Objective 4 | Make memory decisions faster

Initial target: at least 30% lower median memory-management time than the LLM memory manager on the same hardware and workload. Keep final-answer p95 latency within 5% of that baseline while meeting the quality target. Measure writes, retrieval, controller work and complete answer time separately, with warm and cold runs identified.

Objective 5 | Keep facts current without losing history

Initial target: at least 90% current-fact accuracy on a held-out set of explicit corrections, and at least 95% write precision on reviewed decisions. Also measure useful facts missed by the gate. Start with reversible deduplication and archival. Add learned merge, compression and forgetting only after their ablations show that quality remains within Objective 2's margin.

Execution order: reproduce baselines; build the prototype; measure cost and quality together; then optimize one memory operation at a time. Pin models and software versions. Keep training data separate from test conversations. Use at least three runs where stochasticity matters and publish raw per-query logs.

Research contribution to test: whether one inexpensive controller can coordinate writing, retrieval and updates under a context budget. MemRouter, SAGE, EGMW and MemWeave already cover important parts of this idea, so novelty must be demonstrated by controlled comparisons and ablations.

3. Explanations

What the measurements mean

Latency: p50 is the median; p95 is the slower-tail response time. A write decision, a complete answer, a batch total and batch-amortized seconds per query are different measurements. They cannot be ranked as if they measured the same operation.

Token efficiency: prompt tokens are input to the answer model; write tokens are spent creating memory; stored-memory tokens describe memory size. Keeping 62% of turns does not imply saving 38% of tokens. For DAMA, total tokens across all models must also be recorded.

Cost reduction: dollars require prices and a defined workload. Fewer prompt tokens can reduce one cost while additional memory calls increase another. A model-size cost proxy, PFLOPs or smaller memory is not a measured dollar saving.

Quality: F1 measures answer overlap; exact match requires the expected answer; judge scores use an evaluator; pass@1 measures one-attempt coding success. A rise from 70% to 75% is +5 percentage points, or +7.1% relative. These measures are not interchangeable.

Baseline: the system against which the paper compares its method. Full context includes the history in the prompt; RAG retrieves selected evidence; a memory manager can also extract and update facts. A gain over full context does not imply a gain over the strongest memory baseline.

Memory quality: write precision asks whether stored items are useful and correct; missed-fact recall checks useful information rejected by the gate; current-fact accuracy checks whether updates replace stale answers. Pollution refers to irrelevant or harmful stored content under the paper's definition.

How reductions were calculated: $(\text{baseline value} - \text{method value}) / \text{baseline value} \times 100$. Rounding may produce small differences from an abstract's headline. NR means evidence is missing, not that the cost or latency is zero.

How the papers inform DAMA

Write-side evidence comes from MemRouter, SAGE and EGMW; retrieval-side evidence comes from MemWeave and SMS-RAG. Memoria and ReMAP inform current facts and personalization. MemoryRepository motivates retention tradeoffs. Gated-Memory Routing concerns collaboration within a task; the memory-fabric review supplies design requirements. None of these results establishes DAMA's performance.

Paper explanations and evidence

The IDs below identify exactly which paper each table row represents. Links point to the online paper or publisher. Table and section locations identify the evidence used.

1. MemRouter

MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents

A learned gate decides whether to store a turn. Its 12M trainable parameters sit on top of a frozen 7B model and an encoder; they are not the total deployed model size. Useful precedent for cheap writes, with higher reported GPU memory.

Evidence: Section 4.3; Appendix B, Tables 4-6. [Open source](#).

2. SAGE

SAGE: A Novelty Gate for Efficient Memory Evolution in Agentic LLMs

The gate handles clear novelty decisions and sends uncertain cases to an LLM. The shown Mem0 experiment reduces write cost but loses temporal-question quality. DAMA should optimize useful recall, not novelty alone.

Evidence: Tables 1, 4 and 5; GPT-4o-mini / Mem0 experiment. [Open source](#).

3. Gated-Memory Routing

Learning What to Retain: Gated-Memory Routing for Efficient Collaboration in Multi-Agent LLM Systems

The controller selects retained reasoning history and coordinates roles, backbones and stopping within a task. Its compute savings inform controller design, but do not demonstrate persistent conversation memory.

Evidence: Tables 1-3; Appendix 9, Tables 4-5; Appendix 10, Table 7. [Open source](#).

4. Memory fabric

A memory fabric for conversational AI agents enabling shared and persistent multiuser memory

This review discusses shared memory, persistence, provenance and access boundaries. Use it for requirements and architecture; it provides no standalone model benchmark or efficiency claim to reproduce.

Evidence: Abstract and Sections 2-4. [Open source](#).

5. Memoria

Memoria: A Scalable Agentic Memory Framework for Personalized Conversational AI

Session summaries and a weighted knowledge graph produce compact retrieved context and favor current facts. The very large prompt reduction comes from two selected LongMemEval categories and excludes offline memory construction from runtime.

Evidence: Sections V-VI; Tables I-II. [Open source](#).

6. MemoryRepository

MemoryRepository for AI NPC

Summarization and forgetting support long NPC conversations. Human-likeness improves in the shown comparison while retrieval and correctness fall, illustrating why DAMA needs separate recall and presentation-quality measures.

Evidence: Table 3, PDF page 12 (printed page 62592); Section VI. [Open source](#).

7. MemWeave

MemWeave: An Adaptive Framework for Long-Term Agent Knowledge Management via Unified Graph and Vector Memory

A DistilBERT router chooses vector retrieval, graph retrieval or both. This is direct prior work for DAMA retrieval routing. Its routing accuracy and token budget must be evaluated alongside index growth and training-data requirements.

Evidence: Author-posted full text; Sections III-IV, results on PDF pages 3-4 and efficiency table on page 5. DOI: 10.1109/CAIBDA70336.2026.11621567. [Open source](#).

8. EGMW

Evidence-Gated Memory Writing for Personalized Large Language Model Agents

Utility, consistency and evidence checks decide whether to append, update or discard a memory. The method spends more work on writes to reduce noisy memory and later retrieval cost; the benefit depends on the read/write mix.

Evidence: Sections III-V; Table I, PDF page 5; overhead analysis. Publisher metadata checked online; detailed values verified in the repository PDF. [Open source](#).

9. SMS-RAG

Why Retrieval is not Enough: Structured Memory Scheduling for Large Language Model Reasoning

The system schedules semantic, graph and episodic evidence for multi-hop reasoning. Its normalized efficiency chart uses ToG 2.0 as the reference, while the large quality gains in our table use Vanilla RAG. These are separate comparisons.

Evidence: Table 1; Figure 4 on PDF page 12; Section 4.6. [Open source](#).

10. ReMAP

Toward Interpretable and Persistent Personalization: A Memory-Augmented Agent Framework for LLM-Based Travel Planning

The travel planner combines user profiles, factual memory and reasoning. Its ablations motivate personalization and updates, but do not isolate memory latency, token efficiency or total cost. Author-reported percentage gains retain their original wording.

Evidence: Section II-A for Qwen3-235B-A22B; evaluation and ablation sections. DOI: 10.1109/ACCESS.2025.3630495. [Open source](#).

Scope: the ten PDFs currently held in dama-referencepapers. Evidence combines online primary papers/publisher records with the local full texts. These are literature findings, not a common-hardware leaderboard or independent replication.